

# Charbel Junior Nour

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## TECHNICAL SKILLS

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**Skills:** JavaScript, React.js, Java, C++, Python, Ruby, Git, AWS, Arduino, RaspberryPI, Beckhoff TwinCAT3, PLC Programming, Control Systems, Structured Text, MATLAB, Simulink, Autodesk Inventor, Jira, Microsoft 365

**Special Interests:** Guitar, Music Production, 3D Printing, Basketball, Skiing, Cycling, Running

## PROFESSIONAL EXPERIENCE

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### Tesla

*Controls Engineer Intern*

Jan. 2024 – May 2024

- Spearheaded the development of an **RFID**-based tool monitoring system that aims to **reduce** tooling maintenance issues by **over 90%**. Integrated **PLC logic** and started the development of a **custom HMI** to track service history and collect tooling lifespan data on **multiple Gigafactory assembly lines**.
- Supported the commissioning of 3 **automation** machines by leveraging **TwinCAT3 PLC** and **Safety PLC logic** for critical components, including **VFDs**, **IPCs**, various sensors, and robotic and pneumatic devices.
- Contributed to multiple **R&D** initiatives by testing with **time-of-flight lasers** and **profilometers** with an aim to **solve critical engineering challenges** to enhance machine performance in production environments.

*Material Planner Intern*

Sep. 2023 – Dec. 2023

- Developed and maintained 3 internal tools that streamlines the part reordering process through analysis of kitting history, conducting comprehensive **gap analyses**, and employing advanced **Excel VBA** techniques with existing company inventory – resulting in a significantly efficient database.
- Utilized these tools to build a software analysis team that focuses on the automation and standardization of many daily processes, resulting in a **reduction of duplicate shipments by 100%**.
- Facilitated international shipments, expedited missing parts, and streamlined kitting to **reduce technician downtime by 20%**, ensuring efficient workflows across Gigafactories and optimizing daily resource allocation.

### Moneris

*Software Engineer Intern*

Jun. 2022 - Apr. 2023

- Developed and sustained an internal tool that increases productivity for **30+ employees**. The tool displays VISA, MasterCard, and AMEX transaction analytics by using **JavaScript** and **React** hooks to build a responsive and compatible UI that can manage dashboards, widgets, and layout customization.
- Improved code efficiency by **50%** by implementing **Redux** and a backend system hosted by **Parse** to reduce boilerplate, implement reducers, utilize dispatch, and **query user database** properties.
- Collaborated with the QA team in semi-weekly **agile** feedback sessions to identify and address platform issues, enhancing the reliability of a proprietary testing tool for payment platform evaluations.

## PROJECTS

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### Smart Curtain System

- Created a solution for sleep deprivation by designing an automated curtain. The app was coded in **React Native** and the backend is hosted on **AWS IoT** to store data for alarm functions and automated sunrise/sunset control.
- Implemented **C++** code to contact backend server and control the stepper motor to roll up/down the curtain.

### Autonomous RC Car

- Implemented self-driving algorithms coded in **C++** and **Python** on an RC car to navigate through obstacles via **LiDAR** data processed through a **NVIDIA Jetson Nano** running on **Linux Ubuntu**.
- Coordinated meetings to discuss progress and to troubleshoot any development and testing issues.

### Wireless Dog Doorbell System

- Designed an affordable wireless doorbell system enabling puppies to alert owners when they need to go outside, aiding in effective potty training.
- Built a **React Native** interface to provide real-time alerts to owners regarding their puppy's needs.

## EDUCATION

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### B. Eng., Electrical Engineering

Expected Apr. 2025

*McMaster University*

Hamilton, ON

- **Relevant Coursework:** Data Structures & Algorithms, Computer Architecture, High Performance Programming, Principles of Programming, Microprocessor Systems, Signals & Systems, Logic Design, Control Systems, Intro to Mechatronics, Power Electronics, Electric Motor Drives, Electromagnetics I & II, Electric Devices & Circuits
- **Extracurriculars:** General Motors EcoCar Challenge PCM Team Member, GDSC 2023 Incubator Challenge